

Related Work: Step-Level Feedback Distillation

1 Related Work

1.1 Process supervision and process reward models

A line of work has established that supervising how a model reasons, step by step is more effective than supervising only whether the final answer is correct. Lightman et al (1) compared the two regimes directly and found that process supervision substantially outperforms outcome supervision on the challenge MATH dataset, releasing the large PRM800K corpus of step level human labels to support the finding. Because human step annotation is expensive. Math-Shepherd (2) showed that process supervision can be constructed automatically, which is assigning a reward to each solution step without manual labels and used the resulting signal both to rerank candidate solutions and to drive step-level reinforcement learning. These process reward models (PRMs) treat each step’s correctness as a **scalar** quantity and deploy the PRM as a frozen scorer or ranker. Our work shares the step level granularity but departs from the scalar view: we treat the teacher step judgment, including its natural language critique, as the object to be distilled rather than a number to be predicted.

1.2 Generative PRMs

A more recent thread makes the verifier itself generative, producing a reasoning trace before it judges. GenPRMs(3) performs explicit chain-of-thought reasoning with code verification before scoring each step, while ThinkPRM(4) fine-tunes a long-COT verifier that articulates a verification rationale for every step and reaches strong ProcessBench performance from only a few thousand process labels. These models are close to ours in spirit they generate critiques rather than emit bare scores - but they are designed as verifiers evaluated at inference time, not as teacher whose score-plus-critique behavior is distilled into a separate, smaller, answer-blind student. Crucially, none of them isolates when privileged (answer-aware) judgment actually helps, which the question we take up.

1.3 Expert–amateur contrast

Our framing descends from work that extracts signal from the gap between a stronger and a weaker model. CLEAR (5) contrasts feedback from a larger "expert" and a smaller "amateur" model and applies the refined difference to iteratively improve outputs across reasoning tasks; it is the direct predecessor of the present project. Light-Reasoner (6) similarly exploits the behavioral divergence between a strong expert and a weak amateur to pinpoint high-value reasoning moments and distill them, which is notably without ground-truth labels, to sharpen reasoning. We retain the ground-truth free constraints at student test time but invert the asymmetry of interest rather than contrasting models of different capacity. We study a teacher with a privileged view, which has access to reference solution while labeling, distilled into a student that is **answer-blind** by construction. The relevant axis for us is information, not size.

1.4 Our position

Against this backdrop, our contribution is twofold. First, we distill a teacher’s step-level score together with its natural language critique into a small, ground-truth-free student that flags which step a solution first goes wrong. A distillation target richer than the scalar reward of standard PRMs and aimed at error localization rather than solution ranking. Second we characterize when privileged supervision is worth anything, and report a privilege x difficulty x richness effect: a teacher’s answer-aware judgment helps the student only when the problem is hard and the privilege is rich. On easy problems (GMS8K) capable teacher already self-verifies, and the privilege gap collapses to roughly zero; on hard problems (MATH) a bare final answer still moves nothing, while a full worked reference solution lifts error recall by a clear margin. To our knowledge, this conditional structure that privilege buys signal precisely where self-verification fails, and only when the privileged information is a reasoning trace rather than a number has not been isolated in the PRM or expert-amateur literature.

2 Results

2.1 Setup and metric

We evaluate a teacher’s ability to localize the first **erroneous step** in a multi-step solution under three supervision conditions that differ only in the privileged information made available to the teacher at labeling time: (i) **no-GT**, an answer-blind condition in which the teacher sees only the problem and the candidate solution; (ii) **+answer**, in which the teacher is additionally given in the final reference answer; and (iii) **solution**, in which the teacher is given the full worked reference solution. Performance is reported as Process Bench-style first-error **F1**, together with error-detection accuracy. The student remains ground-truth-free at test time throughout; privilege is a property of

the teacher’s labeling pass, not of the deployed evaluator. Reported deltas are taken against the no-GT condition, which serves as the unprivileged reference point.

2.2 Teacher selection

Before probing the effect of privilege, we fix the teacher by an upper-bound bake-off in which every candidate is granted ground-truth access (the with-GT ceiling, $N=30$). The official Gemma-class checkpoint attains the strongest first-error F_1 (0.909) with perfect format compliance (zero parse failures) and the highest throughput among the candidates; a comparably-sized Qwen variant follows closely (F_1 0.873). Notably, a reasoning-tuned candidate collapses under the structured-output protocol, failing to parse on the large majority of instances (parse-failure rate 0.846, F_1 0.000). This is a reminder that verbal reasoning fluency does not by itself confer reliable structured step adjudication. We therefore adopt the Gemma checkpoint as the primary teacher and retain the Qwen variant for the cross-family replication reported below

Teacher	F_1	correct_acc	error_acc	parse_fail
Gemma-4-26b (primary)	0.909	1.000	0.833	0.000
Qwen-27B	0.873	0.917	0.833	0.000
Qwen-35B-A3B	0.839	1.000	0.722	0.019
Qwen-4B-Thinking	0.000	1.000	0.000	0.846

Table 1: Teacher bake-off, with-GT ceiling ($N = 30$).

2.3 The privilege sweet spot

Our central finding is that privileged supervision is not uniformly beneficial: it helps only when the problem is hard enough that the teacher cannot self-verify and tractable enough that the teacher can still exploit reference. The solution-privilege gap is non-significant on easy problems (GSM8K, -0.05, n.s), opens to a significant $0.05F_1$ on hard-but-tractable problems (MATH, 95% CI [0.01, 0.09]), and is non-significant again on the hardest tier (OlympiadBench, -0.03, n.s). The three points trace an inverted-U in difficulty, but we do not lean on the two non-significant endpoints to assert a curve; the curvature is established by mechanism in §2.5, where the rescue rate declines monotonically with reference length. The reading the data support is conjunctive: privilege pays off only where the teacher both needs the reference and can use it.

Tier	Benchmark	solution ΔF_1	Verdict
Easy	GSM8K	-0.05 (n.s.)	saturated; teacher self-verifies
Hard	MATH	+0.05 [0.01, 0.09]	privilege helps
Hardest	OlympiadBench	≈ 0 (n.s.)	reference too long to exploit

Table 2: The privilege sweet spot across difficulty tiers (Gemma teacher). The solution gap rises then falls—an inverted-U, not a monotone trend. MATH is the $N=400$ definitive run (95% CI shown); GSM8K $N=50$; OlympiadBench OE-only, verified.

The two measured endpoints are worth seeing in full. On GSM8K (Table 3, neither the bare answer nor the full solution improves an already-saturated judgment; the teacher is, in effect, its own oracle, and external privilege is redundant. On MATH (Table 4, where unaided self-verification degrades, the full worked solution lifts first-error F_1 from 0.726 to 0.777 (+0.05, 95% CI [0.01, 0.09], significant) and error recall from 0.608 to 0.661. The smaller preliminary MATH run ($N=50$) points the same way with a larger nominal gap ($\sim +0.13 F_1$). OlympiadBench, harder still, returns the gap to ≈ 0 : there the references are too long for the teacher to follow, so the privilege it is handed is not longer usable. The same inverted-U faintly reappears *within* MATH by difficulty level (strongest near the middle, weakest at the easiest and hardest level), corroborating that tractability is the moderator; but the load-bearing evidence is the three-tier contrast, not the noisier per-level split.

GSM8K (Gemma, $N=50$)	F_1	err_acc
NO-GT	0.889	0.800
+ANSWER	0.856	0.800
+SOLUTION	0.837	0.720
Δ (solution)	-0.052	-0.080

Table 3: Easy: saturated.

MATH (Gemma, $N=400$)	F_1	err_acc
NO-GT	0.726	0.608
+ANSWER	0.726	0.608
+SOLUTION	0.777	0.661
Δ (solution)	+0.051	+0.053

Table 4: Hard: the sweet spot. 95% CI [0.01, 0.09].

2.4 Richness, not mere access

The MATH panel isolates a second, sharper effect. The +ANSWER condition, which is a correct final answer but no reasoning, is indistinguishable from the answer-blind baseline: the two rows are identical and the bare answer flips *zero* predictions (measured at $N=150$; we verified the labeling pipeline is correctly wired, so the null effect is behavioral, but not artifact). Only the full worked solution moves the metric. The operative variable is therefore not whether the teacher is privileged but whether the privilege is *rich*: a reference reasoning trace supplies the step-by-step scaffold needed to localize an error, whereas a final answer, which the teacher cannot back-propagate into the faulty step, supplies nothing it can act on.

2.5 Mechanism: privilege rescues self-verification failures

The sweet spot is not a raw difficulty effect; it is a *rescue of self verification failures*, and decomposes into two gates ($N=400$). **Gate 1 (the teacher must need help)**. Of 227 problems containing an error, the NO-GT teacher misses 89. Adding the worked solution rescues 29 of those (33%), while breaking only 17 of the 138 it had already caught (12%), which is the benefit concentrates exactly where self-verification fails. **Gate 2 (the teacher must be able to use it)**. The rescue rate fails monotonically with the length of the reference solution (short 0.37 $\rightarrow$ mid 0.33 $\rightarrow$ long 0.28), which is the mechanism behind the collapse at the hardest tier. Together the gates identify the moderator as self-verification-failure $\times$ tractability, precisely the inverted-U observed across and within datasets.

Illustrate flip cases. The 29 rescued problems (NO-GT miss $\rightarrow$ +SOLUTION catch) illustrate *what* the worked solution lets the teacher see. We group them by error type and show representative cases in Table ??; there are illustrative of the rescue mechanism above, not independent evidence.

Error type	Why NO-GT passed it	Why +SOLUTION flagged it
Unjustified algebraic step	[step] — the manipulation looked locally valid	[the reference takes a different, justified route]
Sign / arithmetic slip	[step] — masked by a plausible continuation	[the reference’s value diverges at that step]
Invalid case split	[step] — each branch seemed self-consistent	[the reference enumerates the complete case set]

Table 5: Representative flip cases by error type (2–3 to be finalized from `results/evidence_pack_n400/per_sample.jsonl`)

2.6 Cross-family replication

To establish that this interaction reflects a property of step-level feedback rather than an idiosyncrasy of the Gemma family, we replicate the MATH probe ($N=150$) with the Qwen-27B teacher (Table 6). The pattern holds: the full worked solution lifts first-error F_1 from 0.653 to 0.735 (+0.08) and error detection from 0.526 to 0.667, while a bare answer yields only a marginal change, consistent with the Gemma result. That the difficulty- and richness-conditioned gain reproduces across two independent model families substantially weakens any single-model-artifact explanation and supports reading the effect as a general characteristic of privileged step-level supervision.

MATH (Qwen-27B, $N=150$)	F_1	err_acc
NO-GT	0.653	0.526
+ANSWER	0.690	0.564
+SOLUTION	0.735	0.667
Δ (solution)	+0.082	+0.141

Table 6: Cross-family replication: the sweet spot is not a Gemma artifact.

2.7 Downstream verifier

The motivating promise of a ground-truth-free step-error localizer is that it can be put to work at *test time*. Given N sampled solutions to a problem, the distilled student PRM scores each and we re-rank, selecting the highest-scored candidate (best-of- N). The question this section will answer is whether such re-ranking beats the standard *majority-vote* baseline on final-answer accuracy—that is, whether privileged step-level supervision, once distilled into a small answer-blind verifier, delivers a usable inference-time gain.

A preliminary Phase-3 run ($N=200$) gives PRM re-rank 32.0 versus majority vote 32.5 (Table 7). More importantly, the re-ranker depends on a Phase-2 student PRM that is *not yet validated*: the initially reported “privilege transfers” gap (F_1 0.197 vs. 0.037) was a fixed-threshold artifact—the NO-GT recall was $\approx 2.6\%$, collapsing that cell’s F_1 —rather than a genuine capability difference. Three things are required before this section can assert anything: (i) a *threshold-free* re-score of the student checkpoints (ROC/PR-AUC), establishing that privileged supervision transfers

to the student by a real margin; (ii) a *symbolic* answer checker for the re-rank accuracy; and (iii) a *larger* N with confidence intervals. We fix the protocol here and defer the result.

Method	final-answer acc.
pass@1 (first candidate)	29.0
PRM re-rank (best-of- N)	32.0
Majority vote (baseline)	32.5
Oracle pass@ N (headroom)	40.5

Table 7: Preliminary best-of- N re-ranking ($N=8$, 200 problems, gemma-2-9b fallback pipeline). The PRM re-ranker (32.0) does *not* beat majority vote (32.5) and is within noise; reported only to fix the protocol. The oracle (40.5) shows substantial headroom remains. Pending a validated student PRM (threshold-free re-score), a symbolic answer checker, and larger N with CIs.

2.8 Does the teacher advantage transfer to the student? (a verified negative)

The teacher-level sweet spot above is the validated result. The natural next question is whether it *distills*: if we train a small (1.5B) student PRM on the privileged teacher’s score-plus-critique labels, does that student inherit a usable advantage over a student trained on no-GT label cross a validated $N=1000$ run, the answer is *no*, which we report this is a clean negative.

References

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